Topic: Span, Linear Independence

Question 1. In the following parts you are given a vector space $\mathcal{V}$ and a subset S of it. Determine if the given subset is linearly dependent or independent.

(a) $\mathcal{V} = \mathbb{R}^3$ and $S = \Big\{(1,-1,2),(1,-2,1),(1,1,4)\Big\}$.

(b) $\mathcal{V} = \mathcal{P}_2(\mathbb{R})$ and $S = \Big\{-x+x^2,\ 3-2x+x^2,\ -2x+x^2\Big\}$.

(c) $\mathcal{V} = \mathcal{P}_2(\mathbb{R})$ and $S = \Big\{x(x-1),\ (x-1)(x-2),\ x(x-2)\Big\}$.

(d) $\mathcal{V} = \mathcal{P}_2(\mathbb{R})$ and $S = \Big\{1,\ 61+80x,\ 33+98x+87x^2\Big\}$.

(e) $\mathcal{V} = \mathcal{F}(\mathbb{R},\mathbb{R})$ and $S = \Big\{f,g,h\Big\}$ where $f(t) = e^{rt}$, $g(t) = e^{st}$, and $h(t) = te^{rt}$ for two fixed distinct real numbers r and s.

Question 2. Prove the following statements.

(a) $\{\boldsymbol{v}\}$ is linearly independent if and only if $\boldsymbol{v}$ is not a zero vector.

(b) $\{\boldsymbol{u},\boldsymbol{v}\}$ is linearly independent if and only if $\boldsymbol{u}$ is not a scalar multiple of $\boldsymbol{v}$ and $\boldsymbol{v}$ is not a scalar multiple of $\boldsymbol{u}$.

(c) If $\mathbf{0} \in S$, then S is not linearly independent.

Question 3. Let $\mathcal{V}$ be a real vector space. Suppose $\boldsymbol{u}$ and $\boldsymbol{v}$ are linearly independent vectors in $\mathcal{V}$. Find condition(s) on real numbers a, b, c and d such that $\{a\boldsymbol{u}+b\boldsymbol{v}, c\boldsymbol{u}+d\boldsymbol{v}\}$ is linearly independent. Is $\{\boldsymbol{u}+\boldsymbol{v}, \boldsymbol{u}-\boldsymbol{v}\}$ linearly independent?

Question 4. Let S be a set of non-zero polynomials in $\mathcal{P}(\mathbb{R})$ with the property that no two of these polynomials have the same degree. Prove that the set S is linearly independent.
Recall that we adopt the convention that the zero polynomial has degree $-\infty$. So, any polynomial with nonnegative degree is *not* the zero polynomial.

Topic: Basis

Question 5. Consider the list $\mathcal{B} = [\boldsymbol{u}_1,\boldsymbol{u}_2,\boldsymbol{u}_3,\boldsymbol{u}_4] \subseteq \mathbb{R}^4$, where

$$\boldsymbol{u}_1 = \begin{bmatrix}1\\1\\1\\1\end{bmatrix},\ \boldsymbol{u}_2 = \begin{bmatrix}0\\1\\1\\1\end{bmatrix},\ \boldsymbol{u}_3 = \begin{bmatrix}0\\0\\1\\1\end{bmatrix},\ \boldsymbol{u}_4 = \begin{bmatrix}0\\0\\0\\1\end{bmatrix}.$$

(a) Explain why $\mathcal{B}$ is a basis for $\mathbb{R}^4$.

(b) Find the unique representation of an arbitrary vector $\boldsymbol{v} \in \mathbb{R}^4$ as a linear combination of vectors in $\mathcal{B}$.

(c) Find the coordinate vector $[\boldsymbol{e}_1]_{\mathcal{B}}$ with respect to the basis $\mathcal{B}$.

Question 6 continued on next page. . .

Question 6. Is the following subset of $\mathcal{M}_{3\times 2}(\mathbb{R})$ linearly dependent or independent?

$$\mathcal{S} = \left\{ \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \right\}$$

If it is linearly dependent, remove some vectors from $\mathcal{S}$ to obtain a basis $\mathcal{B}$ for $\operatorname{span}(\mathcal{S})$.

Question 7. Consider the set $\mathcal{S} = \{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3, \boldsymbol{u}_4, \boldsymbol{u}_5\}$ where

$$\boldsymbol{u}_1 = (2, -3, 1),\ \boldsymbol{u}_2 = (1, 4, -2),\ \boldsymbol{u}_3 = (-8, 12, -4),\ \boldsymbol{u}_4 = (1, 37, -17),\ \boldsymbol{u}_5 = (-3, -4, 8)$$

(a) Form a matrix $\boldsymbol{A}$ whose ith column is the column vector $\boldsymbol{u}_i^T$. Find the reduced row-echelon form of $\boldsymbol{A}$.

(b) Use part (a) to explain why $\mathbb{R}^3 = \operatorname{span}(\mathcal{S})$.

(c) Use part (a) to reduce the set $\mathcal{S}$ to a basis $\mathcal{B}$ for $\mathbb{R}^3$, and express other vectors not in $\mathcal{B}$ in terms of basis vectors in $\mathcal{B}$.

Question 8. Determine whether the following list $\mathcal{S}$ is a basis for $\mathcal{V}$.

(a) $\mathcal{S} = [(-1, 3, 1, 0), (2, -4, 0, -3), (-3, 0, 8, 2), (0, 0, 1, -2)], \quad \mathcal{V} = \mathbb{R}^4$.

(b) $\mathcal{S} = [(3, 1, -1, 0), (2, 0, -3, 4), (1, 0, 0, 5), (3, 2, 5, 1), (2, 0, 1, 9)], \quad \mathcal{V} = \mathbb{R}^4$.

(c) $\mathcal{S} = [1 + 2x - x^2, 4 - 2x + x^2, -1 + 18x - 9x^2], \quad \mathcal{V} = \mathcal{P}_2(\mathbb{R})$.

(d) $\mathcal{S} = [\pi, x - \sqrt{2}x^3, 1102 - 3x^2, -6x, x^2 + 2019x^3], \quad \mathcal{V} = \mathcal{P}_3(\mathbb{R})$.

Question 9. Find a basis for each of the following subspaces of $\mathcal{M}_{4\times 4}(\mathbb{R})$, and determine its dimension.

(a) $\boldsymbol{W}$ is the subspace of diagonal matrices.

(b) $\boldsymbol{W}$ is the subspace of symmetric matrices.

Question 10. Consider the subspace $\mathcal{W} \subseteq \mathcal{P}_4(\mathbb{R})$ where

$$\mathcal{W} = \{p(x) \in \mathcal{P}_4(\mathbb{R}) : \ p(0) = 0 \text{ and } p(2) = 0\}.$$

Find a basis $\mathcal{B}$ for $\mathcal{W}$. Justify your answer.

Question 11. Suppose $\mathcal{V}$ is a real vector space with a basis $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$. Let $\mathcal{S} = \{\boldsymbol{v}_1 - 2\boldsymbol{v}_2 + 4\boldsymbol{v}_3, 2\boldsymbol{v}_1 + \boldsymbol{v}_3, -2\boldsymbol{v}_1 + 7\boldsymbol{v}_2 - \boldsymbol{v}_3\}$. Is $\mathcal{S}$ a basis for $\mathcal{V}$?

Question 12. Suppose $\mathcal{V}$ is a vector space and $\mathcal{S} = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\}$ be a set of vectors in $\mathcal{V}$, where $k \geq 2$. Let $\mathcal{S}^* = \{\boldsymbol{v}_1 - \boldsymbol{v}_2, \boldsymbol{v}_2 - \boldsymbol{v}_3, \ldots, \boldsymbol{v}_{k-1} - \boldsymbol{v}_k, \boldsymbol{v}_k\}$.

(a) Prove that $\operatorname{span}(\mathcal{S}) = \operatorname{span}(\mathcal{S}^*)$.

(b) If $\mathcal{S}$ is a basis of $\mathcal{V}$, is $\mathcal{S}^*$ a basis of $\mathcal{V}$?

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Question 13. Extend the linearly independent set $\mathcal{S}$ to a basis $\mathcal{B}$ for $\mathcal{V}$, and determine the coordinate vector $[\boldsymbol{v}]_{\mathcal{B}}$.

(a) $\mathcal{V} = \mathbb{R}^4$, $\mathcal{S} = \{(2,0,3,0),(1,1,0,2)\}$ and $\boldsymbol{v} = (1,1,1,1)$.

(b) $\mathcal{V} = \mathcal{M}_{2\times 2}(\mathbb{R})$, $\mathcal{S} = \left\{ \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 0 \\ -1 & 0 \end{bmatrix} \right\}$, and $\boldsymbol{v} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$.

(c) $\mathcal{V} = \mathcal{P}_3(\mathbb{R})$, $\mathcal{S} = \{1 - x + x^2 - x^3, -2x + 3x^2, 3 - x\}$ and $\boldsymbol{v} = x^2$.

Question 14. Let $f(x)$ be a fixed polynomial of degree n in $\mathcal{P}_n(\mathbb{R})$.
Prove that for every $g(x) \in \mathcal{P}_n(\mathbb{R})$, there exist scalars $\lambda_0, \lambda_1, \ldots, \lambda_n \in \mathbb{R}$, such that

$$g(x) = \lambda_0 f(x) + \lambda_1 f'(x) + \cdots + \lambda_k f^{(k)}(x) + \cdots + \lambda_n f^{(n)}(x),$$

where $f^{(k)}(x)$ denotes the kth derivative of $f(x)$.
Hint: What can you say about the degrees of $f(x), f'(x), f^{(2)}(x), \ldots, f^{(n)}(x)$?

Question 15. Fix positive integer n and some point $a \in \mathbb{R}$. Consider the subset $\mathcal{W} \subset \mathcal{P}_n(\mathbb{R})$ where $\mathcal{W}$ consists of the polynomials $p(x)$ such that $p(a) = 0$.

(a) Verify that $\mathcal{W}$ is a subspace of $\mathcal{P}_n(\mathbb{R})$.

(b) Find a basis for $\mathcal{W}$. (Hint: Suppose that $p(x) \in \mathcal{W}$. What can you say about $p(x)/(x-a)$?)

(c) What is the dimension of $\mathcal{W}$?

Question 16. Suppose $\mathcal{V}$ is a finite dimensional space with $\dim(\mathcal{V}) = n$. Let $\mathcal{U}$ and $\mathcal{W}$ be subspaces of $\mathcal{V}$. Recall the sum $\mathcal{U} + \mathcal{W}$ of subspaces $\mathcal{U}$ and $\mathcal{W}$ is defined as

$$\mathcal{U} + \mathcal{W} = \{\boldsymbol{u} + \boldsymbol{w} \in \mathcal{V} : \boldsymbol{u} \in \mathcal{U} \text{ and } \boldsymbol{w} \in \mathcal{W}\}.$$

Recall that we have proved that $\mathcal{U} + \mathcal{W}$ is a subspace of $\mathcal{V}$.

(a) Prove that $\dim(\mathcal{U} + \mathcal{W}) = \dim(\mathcal{U}) + \dim(\mathcal{W}) - \dim(\mathcal{U} \cap \mathcal{W})$.

(b) Is the following statement true? (Use part (a) of course.)
If $\dim(\mathcal{U}) + \dim(\mathcal{W}) > n$, then $\dim(\mathcal{U} \cap \mathcal{W}) > 0$.